

Serum polycyclic aromatic hydrocarbons among children with and without asthma: correlation to environmental and dietary factors.



BACKGROUND: Children from low-income families may be subject to high exposures to polycyclic aromatic hydrocarbons (PAH) which can lead to respiratory disorders. This study aims to establish methods for assessing total PAH exposure of asthmatic and non-asthmatic children from low-income families; to estimate serum PAH concentrations of these children, and to estimate the relative importance of the environmental pathways for PAH exposure. MATERIALS AND METHODS: A total of 75 (61 asthmatic, 14 non-asthmatic) Saudi children 15 years old and below were included to participate in this cross-sectional study. Each participant answered a generalized questionnaire with dietary questions. Serum PAH were measured using HPLC with UV detection. RESULTS: Serum naphthalene and pyrene were significantly elevated among asthmatic children (p-values = 0.007 and 0.01, respectively). Serum acenaphthylene, fluorine and 1,2-benzanthracene, on the other hand, were significantly higher among non-asthmatics (p-values = 0.001, 0.04 and 0.03, respectively). There was a significant correlation between the presence of a smoker in the family and serum concentrations of carbazole, pyrene, 1,2-benzanthracene and benzacephenanthrylene ($R = 0.37, 0.45, 0.43, 0.33$; p-values = 0.01, 0.0002, 0.003 and 0.025, respectively). Significant correlations were elicited between daily meat intake and serum levels of acenaphthylene, benzopyrene and 1,2-benzanthracene ($R = 0.27, 0.27, 0.33$; p-values = 0.02 and < 0.001 , respectively). CONCLUSION: Among the children, serum PAH were significantly correlated to meat intake as well as presence of smokers at home. Public health awareness should be enhanced by educating parents to take certain precautions at home, such as preventing indoor smoking and reducing the intake of grilled and smoked meat by children so as to decrease their exposure to carcinogenic PAH.